Beyond Variance: Asymmetric Labor Income Risk and Portfolio Entry

Tai Lo Yeung
Università della Svizzera italiana

Job Market Paper · October 2026

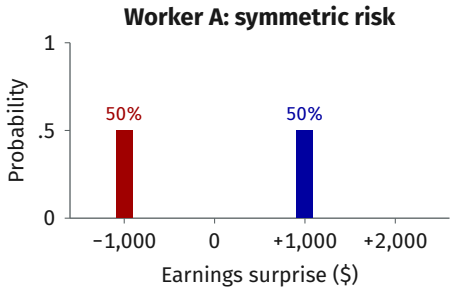
Can earnings opportunities help people start investing?

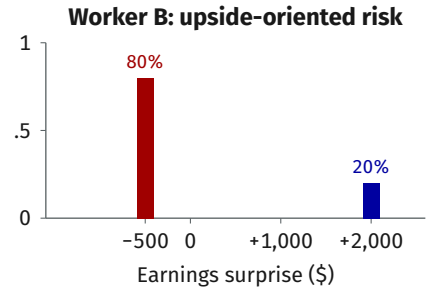
- **Why entry matters.** Starting to own stocks gives households another way to build wealth.
- **The familiar concern.** Uncertain pay can make taking investment risk harder.
- **Variance remains a common summary.** It measures the size of earnings fluctuations, treating gains and losses symmetrically.

  Recent examples: d'Astous & Shore (JFE, 2024); Bacher (RFS, 2025).

Research question: Does the balance of upside and downside earnings variation predict who becomes an investor?

Same variance. Same decision to invest?





Same mean surprise: \$0 Same variance: $\$1{,}000^2$

B has smaller frequent losses and larger rare gains.

More upside predicts more stock-market entry

U.S. Survey of Income and Program Participation (SIPP)
27,419 annual transitions: 2019–2023 baselines, 2020–2024 outcomes.

- **Finding.** Workers in groups with more upside-oriented earnings variation are more likely to start owning stocks or mutual funds outside retirement accounts.
- **Economic size.** A one-standard-deviation increase in earnings composition across worker groups predicts **1.60 percentage points more entry**.

  About **16 more entrants per 1,000** initial nonholders.
  Annual entry rate in this sample: 8.82%.
- **Model mechanism.** Earnings opportunities affect both the money people receive and how much they choose to save before investing.

▶ Economic contribution ▶ Coefficient units

Economic contribution

The main contribution is to establish the economic importance of upside earnings opportunities for stock market entry.

Separating favorable from adverse variation reveals substantial, opposing entry associations that an imprecise linear variance–entry relationship conceals.

Three connected questions

1. Which features of earnings risk matter for portfolios?
2. How do households become investors?
3. What should portfolio models explain beyond average holdings?

▶ Measurement comparisons ▶ Closest studies ▶ Related work in progress

Which features of earnings risk matter for portfolios?

- **The amount of risk remains central.**
  d'Astous & Shore (JFE, 2024); Bacher (RFS, 2025)
  Earnings variance predicts participation and risky shares. Life-cycle models use persistent and temporary income-shock variances.
- **The shape of risk also matters.**
  Catherine, Sodini & Zhang (JF, 2024); Shen (Management Science, 2024)
  Downside earnings risk in poor stock markets matters for stockholding and saving.
- **Earnings risk over a career.**
  Gálvez & Paz-Pardo (JFE, 2026)
  Richer earnings dynamics change participation, risky shares and estimated risk aversion.

Here: More upside relative to downside predicts entry at similar mean growth, total variance and measured resources.

▶ Earlier evidence ▶ Closest studies ▶ Related work in progress

How do earnings opportunities translate into market entry?

- **Resources and ownership transitions.**
  Briggs et al. (JFE, 2021); Bonaparte et al. (Management Science, 2025)
  Lottery windfalls raise entry among nonholders. Income and wealth changes also help explain entry and exit.

- **This paper measures earnings opportunities before entry.**
  Earlier earnings distributions, measured in a separate survey, predict entry and entry followed by later ownership.

- **The model tests the same entry relationship.**
  Entry–risk coefficients are excluded from fitting. The model separates changes in resources along earnings paths from changes in saving and investment decisions.

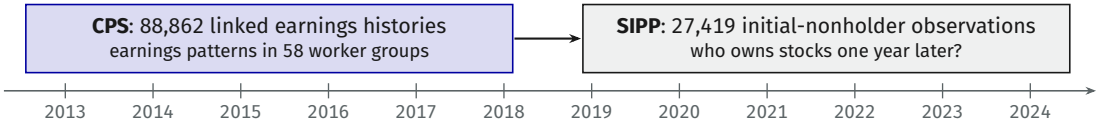
▶ Closest empirical studies ▶ Model comparison

How will we answer the question?

1. **Measure:** where does earnings variation come from?
2. **Evidence:** who starts investing, and who keeps holding?
3. **Checks:** earnings measurement, sorting and reporting
4. **Model:** how do earnings paths and saving decisions shape entry?

Two public surveys connect earnings risk to investment

- **Current Population Survey (CPS).** Year-to-year earnings changes measure risk in 58 worker groups, defined by industry, education and age.
- **Survey of Income and Program Participation (SIPP).** Repeated household interviews track wealth and stock ownership.



Assign each SIPP observation its group's earlier CPS earnings measures. These are separate public-use U.S. surveys.

Entry: no stocks or mutual funds today, ownership next year, outside retirement accounts.

▶ Sample and units ▶ Why earlier measurement?

Split earnings variance into two parts

Start with age-adjusted earnings growth in a worker group.

Measure each outcome relative to that group's mean.

$$V^- = \mathbb{E}[(\epsilon - \mu)^2 \mathbf{1}\{\epsilon < \mu\}], \qquad V^+ = \mathbb{E}[(\epsilon - \mu)^2 \mathbf{1}\{\epsilon > \mu\}].$$

Amount of variation

$$V = V^+ + V^-$$

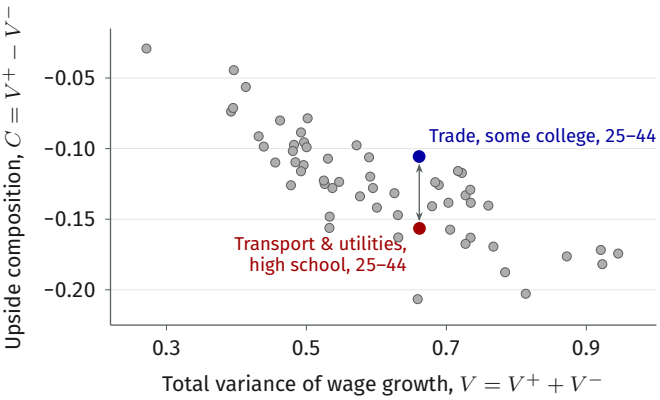
Balance of gains and losses

$$C = V^+ - V^-$$

At fixed V, higher C shifts variation from the lower to the upper side.

▶ Earnings construction ▶ Numerical example ▶ Relation to skewness

Variation across earnings cells



58 earnings cells

Highlighted pair:
Variance ≈ 0.66
Composition gap ≈ 1.5 source SD

All cells: $C < 0$
Relative upside measure

$\text{corr}(C, V) = -0.81.$

Entry at similar total variance and resources

$$H_{i,t+1} = \alpha + \beta \frac{C_c - \bar{C}}{s_C} + \text{controls}, \qquad H_{it} = 0.$$

Outcome	Stock/mutual-fund entry next year: 0 or 1
Total earnings variance	V and $\sqrt{V}$: the amount of earnings variation
Mean earnings growth	Mean growth and its square
Resources	Initial assets, earnings and nonpositive-earnings indicator
Demographics	Age, age squared and sex
Fixed effects	Industry, education, age band and outcome year

Weighted linear probability model; industry–education clustered inference.

▶ Full specification ▶ Why education and sex?

Earnings composition and next-year entry

Composition and entry	Estimate
Entry difference per source SD	$+1.60$ pp
Clustered standard error	0.51 pp
Mean annual entry	8.82%
Initial nonholder transitions	27,419

About 16 more entrants per 1,000 initial nonholders.

Roughly 18% of the mean entry rate, at similar dispersion and resources.

▶ All earnings measures ▶ Coefficient units ▶ Published magnitudes

Nonlinear variance and the additional composition association

Risk terms	(1) V	(2) $V, \sqrt{V}$	(3) $C, V, \sqrt{V}$
Composition C	—	—	$+1.601^{***}$
			(0.506)
Variance V	-0.457	-11.839^{***}	-15.380^{***}
	(0.584)	(3.845)	(3.505)
Standard deviation $\sqrt{V}$	—	$+11.457^{***}$	$+15.762^{***}$
		(3.689)	(3.456)
Nonholder transitions	27,419	27,419	27,419

Both variance terms remain significant with composition. Their combined local slope can be small near the curve's peak.

Same sample, weights and other controls. Direct coefficients in pp per own source SD; clustered SEs in parentheses. $^{***}p < 0.01$.

▶ Interpreting the nonlinear relationship ▶ Composition estimate

What the next checks address

Concern	Comparison
Earnings definition or zero wages	Four earnings constructions, same SIPP entry sample
Group differences and changing conditions	Richer group and time controls
Ownership reporting or household changes	Reported, self-reported, stable households
Another survey and period?	PSID entry, full sample and exploratory age groups
Only temporary entry?	Entry together with later ownership; longer followup cohorts
Do individuals' beliefs line up?	SCE subjective expectations and retirement portfolios

▶ Why the samples differ

Checks tied to specific concerns

Each check changes the baseline separately.

Concern	Specification	Estimate, pp (SE)	Nonholders
	Prospective baseline	1.646^{**} (0.599)	24,235
Group-year conditions	Industry $\times$ year; education $\times$ year	1.663^{**} (0.619)	24,235
Persistent group sorting	Instead: industry $\times$ education effects	1.865^{**} (0.875)	24,235
Reporting and household changes	Reported, self-reported, stable households	2.557^{***} (0.555)	11,229
Realized earnings: a separate common sample of 23,261 nonholders			
	Baseline	1.518^{**} (0.587)	23,261
	Add annual earnings level and change	1.574^{**} (0.586)	23,261

▶ Sorting ▶ Samples

Evidence from the PSID

Biennial entry among initial nonholders; baseline years 1999–2017.

Age range	Composition (pp)	SE
Full sample: 25–65	-0.18	0.39
Exploratory age split		
25–44	2.42^{**}	1.11
45–65	-0.98	1.01

No clear full-sample association; a positive estimate among younger workers.

PSID uses different households. Some outcomes occur before the CPS earnings data, so this is not the same timing design as SIPP.

▶ Later-period age comparison ▶ Period differences

Entry followed by later ownership

Within each column: the same nonholders, two outcomes.

Follow-up cohort	One later year	Two later years
Entry alone (pp)	2.09^{***} (0.70)	3.31^{**} (1.43)
Entry + later ownership (pp)	0.98^{*} (0.50)	1.89^{**} (0.79)
Sustained outcome $= 1$	$0, 1, 1$	$0, 1, 1, 1$
Initial nonholder transitions	9,236	3,254

The sample includes all initial nonholders with the required annual follow-up.

Other observed sequences have sustained outcome 0.

▶ Entry and interruption ▶ Sample and sequences

Upside beliefs and risky reallocation

Survey of Consumer Expectations (SCE), retirement accounts

Belief measure	Reallocation	Equity share
Upside-minus-downside tilt	$\mathbf{0.028}^{**}$	-0.576
	(0.013)	(0.664)
Sum of semideviations	-0.017	$\mathbf{-3.805}^{***}$
	(0.018)	(1.011)
Respondents	2,136	3,332
Controls	Expected earnings growth, age, education and income	
Fixed effects / SEs	State and module year / 50 state clusters	

Reallocation score: $+1$ riskier, -1 safer, 0 unchanged. Equity share: percentage points.

Earnings composition and participation decisions

- **Main result:** more upside-oriented variation predicts more next-year entry at similar dispersion and resources.
- **Later ownership:** the association also appears for entry followed by ownership at later annual observations.
- **Economic question:** how does the earnings environment relate to a nonholder's subsequent investment decision?

The model tests a familiar resource mechanism against this entry pattern.

▶ Entry and retention

A familiar mechanism: earnings can finance entry

- **Standard household choice:** consume, save, and pay a cost if holding stocks is worthwhile.
- **Why composition can matter:** equal means and variances can conceal different chances of earning enough to invest.
- **What this paper tests:** does that mechanism produce the observed entry association under measured earnings environments?

The cells supply earnings differences. The saving and participation mechanism is shared across households.

The earnings–entry coefficients are never fitting targets.

▶ Related models ▶ Alternative access model

Two dates: when is paying the fee worthwhile?

1. **Today:** earnings arrive. Cash is $m = w + Y$.
2. **Today:** choose consumption and saving. Pay fee κ to invest part of saving in stocks.
3. **Tomorrow:** consume the return on saving. Stocks offer an expected return advantage, with risk.

Optimal choice: participate when cash exceeds a threshold.

$$\Pr(\text{entry}) = \Pr\left(Y \geq \underbrace{m^* - w}_{\text{cash shortfall}}\right).$$

A fixed fee is easier to justify when more money can be invested.

▶ Equal-variance example ▶ Optimal threshold

Same variance, different chances of reaching the threshold

Earnings innovation	Symmetric	More upside
Possible loss	$-\$1{,}000$ (50%)	$-\$500$ (80%)
Possible gain	$+\$1{,}000$ (50%)	$+\$2{,}000$ (20%)
Mean / standard deviation	\$0 / \$1,000	\$0 / \$1,000
Entry if \$1,500 short	0%	20%
Entry if \$500 short	50%	20%

The gain matters when it carries a household across the threshold.

The sign depends on where households start. Cells are not needed for this mechanism.

Illustrative dollar innovations around common expected earnings, not estimated responses.

◀ Back ▶ Derivation

Can a workhorse model reproduce the entry association?

1. Estimate earnings processes from CPS and PSID earnings moments.
2. Fit five financial parameters to eight SIPP financial outcomes.
3. Solve the same saving and participation problem under cell and pooled earnings dynamics.
4. Apply the empirical entry regression to those predictions.

Question: how much of the observed composition association do measured earnings environments generate?

Related risk-regression comparison: Gálvez & Paz-Pardo (2026).

▶ Initial states ▶ Published precedents

What households know when they choose

Before choice	Current earnings and asset returns have arrived. Cash M_t, earnings state, access and participation cost are known.
Choice	Consume, save, and decide whether to hold stocks. Choose the risky share if participating.
After choice	Next earnings and returns remain uncertain. Their distributions enter expected continuation utility.

$$c_t + A_t + D_t K_t = M_t, \qquad M_{t+1} = A_t R_{t+1}(\alpha_t) + Y_{t+1}.$$

Y: resources after taxes and own retirement contributions.

Prior holders have access. Nonholders receive an opportunity with probability q. Holding pays K_t; participation requires a funded position.

▶ Recursive problem ▶ Constraints ▶ Realizations and expectations

What determines a worker's earnings?

While employed, log wages combine a career profile and two shocks.

$$\log W_{gt} = \underbrace{f_g(t)}_{\text{predictable profile}} + \underbrace{\eta_t}_{\text{persistent state}} + \underbrace{\varepsilon_{gt}}_{\text{temporary shock}} .$$

Career profile	Typical pay at that age, education and industry
Persistent state	Small ordinary innovations plus occasional promotions or earnings losses that carry forward
Temporary shock	Variation in current pay that does not carry forward

A separate employment transition determines whether wages are zero.

Disposable income also reflects taxes, own retirement contributions and resources outside wage work.

▶ Where asymmetry enters ▶ Exact earnings equation

How do 58 measured cells enter the model?

An earnings law gives probabilities of future pay and wage work.

Shared foundation	Two career processes: college and noncollege
30 career groups	Industry $\times$ education, with group wage levels
58 observed cells	Separate adjustments before and after age 45 within those groups

Cell adjustments change earnings growth, departures from wage work, transitory variation and asymmetry.

Everyone shares preferences, participation costs and opportunity probability.

Two unsupported group–age combinations use the other age band's adjustment.

▶ Exact mapping ▶ Why 58, rather than 580? ▶ Four adjustments

Four adjustments connect each cell to its earnings data

Cell adjustment	What it changes
Departure risk	Probability of leaving wage work
Growth	The predictable earnings path with age
Shock scale	The size of transitory variation and reporting error
Asymmetry tilt a_c	Relative weights on favorable and adverse persistent and temporary shocks

Fit all four adjustments jointly to four CPS earnings moments:

zero-wage frequency, mean growth, upper partial variance and lower partial variance.

PSID longer-horizon moments discipline the shared career process. Stock-market entry coefficients never enter this calibration.

▶ How the tilt works ▶ Exact cell mapping

Gains and losses need not be mirror images

Persistent earnings combine ordinary innovations with occasional jumps.

$$\eta_{t+1} = \rho\eta_t + \zeta_{t+1} + \underbrace{J^+ I^+_{t+1}}_{\text{promotion}} - \underbrace{J^- I^-_{t+1}}_{\text{earnings loss}} - m.$$

ρ: persistence; ζ: small normal innovation; m: centering drift.

Promotion and loss are mutually exclusive. Their sizes and probabilities can differ.

A positive cell tilt a_c shifts relative weight toward favorable shocks.

$$\text{Promotion weight } \propto e^{a_c}, \qquad \text{Loss weight } \propto e^{-a_c}.$$

Temporary pay shocks follow a mixture distribution. The same tilt shifts their relative weights toward the upper tail.

▶ Exact persistent tilt ▶ Temporary-shock tilt

Calibration and parameter estimates

Calibration and data inputs

Inputs held fixed	Value / source
Safe gross real return	1.02
Ordinary stock gross mean / SD	1.06 / 0.18
Annual crash probability	1%
Crash net real return	-48.5%
EIS ψ	0.5
Earnings laws / resources	CPS / PSID
Observed initial states	SIPP

Estimated financial parameters

Parameter	Estimate
Risk aversion γ	8.801
Discount factor β	0.811
Recurring mean κ_R	\$524
Log-cost dispersion σ_κ	0.944
Opportunity probability q	0.176

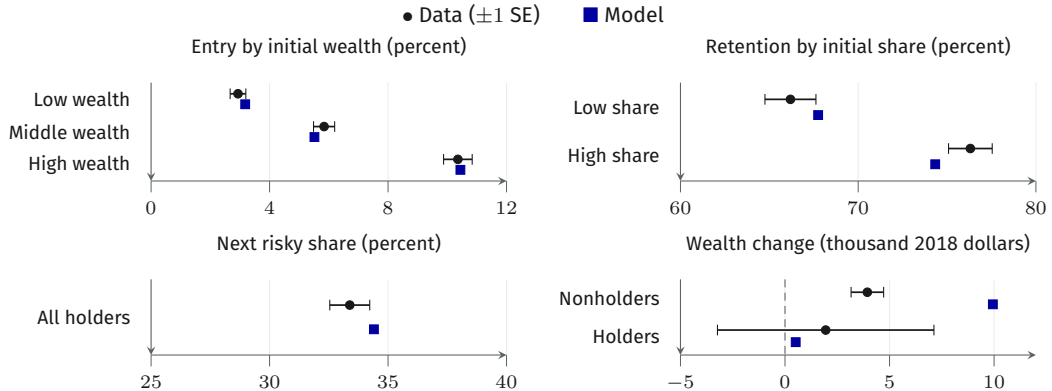
Joint simulated method of moments (SMM): five common parameters, eight financial targets.

Preferences: Epstein–Zin. **Calibrated:** returns and EIS. **Estimated separately:** earnings laws. **From data:** resource profiles and initial household states.

▶ SMM targets and weighting ▶ Financial fit ▶ Criterion and numerical checks

Financial targets: data and model

Eight financial moments; five estimated parameters



Main discrepancy: excessive wealth growth among initial nonholders.

◀ Back ▶ Full target table ▶ Index

How does the earnings law change a decision?

Realized income	Earnings arrive and become cash available for consumption, saving and participation fees.
Future income prospects	Probabilities of future pay and wage work change the value of saving and holding stocks today.

$$c_t + A_t + D_t K_t = M_t, \qquad M_{t+1} = A_t R_{t+1}(\alpha_t) + Y_{t+1}.$$

The decision rule is the optimal plan for each cash and earnings state.

It chooses consumption, saving, participation and the stock share, using both current resources and future earnings probabilities.

At the same cash, different prospects can change the plan. Preferences, costs and financial return expectations remain common.

▶ Continuation values ▶ Law and rule accounting

Two-period illustration: same mean and variance

Earnings today and next year follow one of these two laws.

All amounts are in thousands of dollars.

	Upside cell (U)	Downside cell (D)
Ordinary outcome	85 with 80% chance	115 with 80% chance
Tail outcome	160 with 20% chance	40 with 20% chance
Mean / standard deviation	100 / 30	100 / 30

U has a rare large gain; D has a rare large loss.

Same log utility and patience; entry costs 1.5. Safe saving earns 0%; stocks return -50% or $+70\%$, each with 50% chance, independently of earnings.

These are illustrative cells, separate from the fitted 58-cell model.

▶ Full assumptions ▶ Same cash, different prospects

Two-period illustration: expectations at the same cash

Give both households 100 in cash today. Only future earnings differ.

Future earnings law	Consume today	Save safely	Enter stocks?
Upside cell (U)	96.96	3.04	No
Downside cell (D)	92.77	7.23	No

The downside household saves 4.19 more as a buffer.

Both expect earnings of 100 tomorrow. A 20% chance of earning only 40 makes extra saving more valuable in D.

This comparison isolates expectations: current resources, mean future earnings, variance and financial opportunities are the same.

All amounts are in thousands of dollars. Uncalibrated optimal choices.

◀ Back ▶ Let earnings arrive ▶ Full assumptions

Two-period illustration: let earnings become cash

Start with no inherited assets. Current and future earnings are independent draws from the same cell law. Amounts are in thousands of dollars.

Current outcome	Chance	Cash	Optimal saving	Enter?
U: ordinary	80%	85	0	No
U: large gain	20%	160	32.22 in stocks	Yes
D: large loss	20%	40	0	No
D: ordinary	80%	115	14.02 safely	No

The large gain finances stock entry despite the fee.

U: 20% enter; mean saving is 6.44. D: 0% enter; mean saving is 11.22.

The group saving gap combines realized-resource and expectation differences. It is not an isolated precautionary-saving effect.

◀ Back ▶ Full assumptions

What changes when we pool earnings dynamics?

	Cell dynamics	Pooled dynamics
Earnings adjustments	Group and age specific	Removed
Career foundation	Same education processes and group wage levels	
Starting point	Same observed households and initial resources	
Preferences and costs	Same fitted financial parameters	
Decisions	Solve optimal choices under each earnings law	

This asks what measured earnings differences add at common parameters.

Pooling changes the full earnings environment. It is not an asymmetry-only shock or a separately fitted benchmark.

▶ Untargeted comparison ▶ Separate composition experiment

Apply the same regression to model entry probabilities

	Data	Model
Outcome	Entry observed: 0 or 1	Predicted entry probability
Households	The same 17,606 initial nonholders	
Right-hand side	Same risk measures, controls, fixed effects and weights	

Average over earnings, opportunities and costs, conditioning each origin on next-year wage work and positive assets.

Compare gross cell predictions with data.

Subtract pooled predictions to describe the contribution of cell earnings laws.

The matched data coefficient is 1.816 pp; the full-sample headline is 1.601 pp.

▶ Why probabilities do not shrink coefficients ▶ Matched sample

Measured earnings differences strengthen the entry association

Composition coefficient: percentage points per source SD

Same 17,606 initial nonholders	Entry association
Observed data	1.816^{***} (SE 0.480)
Model with pooled earnings dynamics	-0.005
Model with cell earnings dynamics	$+0.241$

Cell dynamics add **0.247 pp** at common financial parameters.

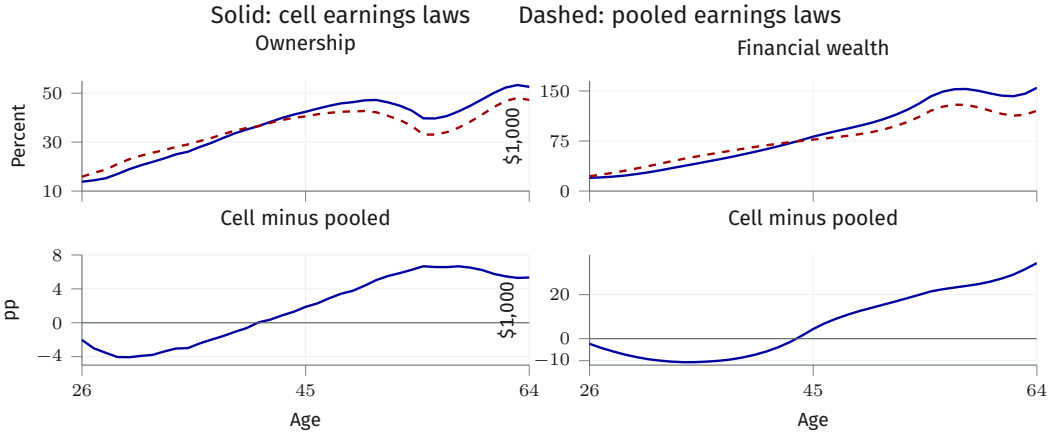
The sign agrees with the data. The magnitude remains much smaller.

Same regression, controls and weights. These coefficients are untargeted.

▶ Resources and decision rules ▶ Full comparison ▶ Alternative model

Ownership and wealth paths

Same 135 age-25 origins; two college groups; simulated paths



An illustration: shift variance toward upside or downside

A separate experiment within the fitted cell earnings laws.

Let U_y and L_y be the upper and lower partial variances of earnings in dollars.

More upside: $(U_y + d, L_y - d)$ More downside: $(U_y - d, L_y + d)$.

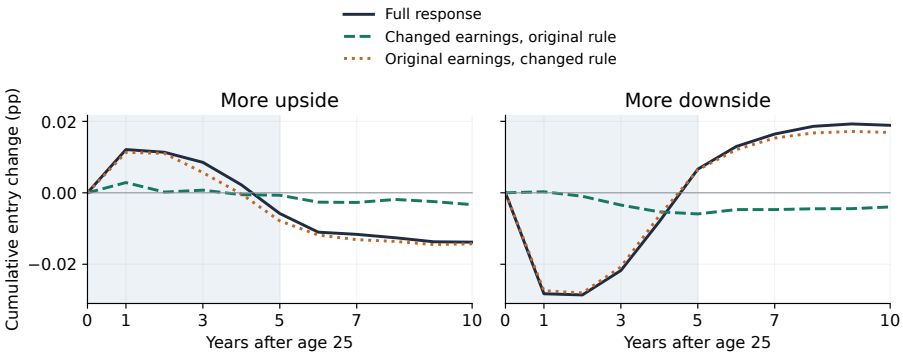
- d transfers variance, not income. Expected disposable earnings and total dollar variance stay fixed.
- **Years 1–5 (ages 26–30):** draw from the changed distribution each year. **Year 6 onward:** baseline. Households know this schedule.
- Parameters, starting states and opportunity/return draws fixed; risk coefficients untargeted.

Follow 395 initial nonholders from age 25 for ten years. The shock illustrates the mechanism; its size is not fitted to the empirical coefficient.

Dollar moments are fixed; growth-rate moments can change.

▶ Entry ▶ Saving at the same cash

More upside can raise entry first, then lower it



The changed saving and investment plan drives most of the later reversal.
Years 1–5: changed earnings distribution. Year 6 onward: baseline. Households know this schedule.

▶ Wealth ▶ Saving at the same cash

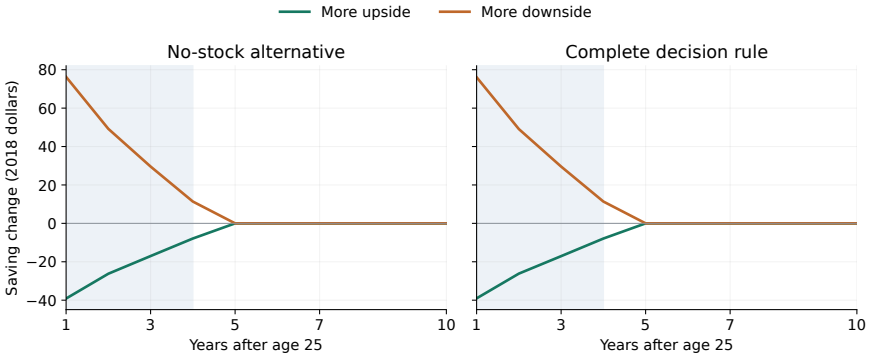
The upside shift reduces accumulated financial wealth



The effect persists through the financial states households carry forward.
From age 30, rules coincide at identical cash and earnings states; past choices still affect wealth.

▶ Same-cash comparison ▶ Consumption and fees

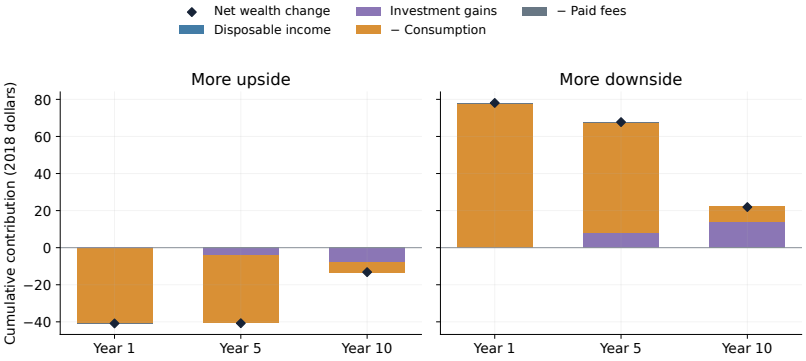
At the same cash, more upside lowers desired saving



First year: saving falls by about $39, even in the no-stock alternative.
Changed future earnings prospects reduce the demand for a financial buffer.

▶ Where the money goes ◀ Back

Higher consumption accounts for the initial wealth decline



First-year upside response: consumption rises $40.81; fees rise $0.03.
More favorable prospects can encourage entry now while reducing the saving that finances entry later.

◀ Back ▶ One year or lasting prospects? ▶ Model assessment

Why five years? Compare three durations

Five years is an illustrative horizon, not an estimated duration.

Duration	Affected ages	What follows
One year	26	Baseline earnings from age 27
Five years	26–30	Baseline earnings from age 31
Remaining working life	26–64	Retirement income unchanged

Same annual shock at shared ages; expected disposable earnings and dollar variance stay fixed.

Same 395 initial nonholders and fixed parameters; risk coefficients remain untargeted.

Households know the schedule. Starting portfolios are given; the first new choice follows year 1 earnings.

◀ Back ▶ Entry comparison ▶ Wealth comparison

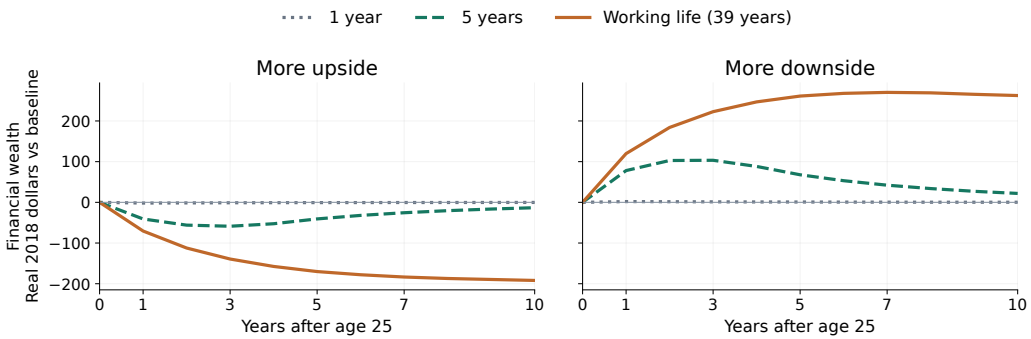
Does the entry response depend on duration?



The five-year reversal depends on how long earnings prospects change.
One-year entry is near zero and grid-sensitive. Cumulative entry under lasting upside fades toward zero. Its later annual first-entry differences are negative.

◀ Back ▶ Wealth ▶ Annual first entry

How does duration change financial accumulation?



Lasting prospects sustain the saving and wealth response.
At year 10, lasting upside lowers wealth by about $192; lasting downside raises it by $262. The five-year gaps are about $13 and $22.

◀ Back ▶ Why wealth and entry differ ▶ All comparisons

Why can wealth fall while cumulative entry barely changes?

- **Lasting prospects keep two forces active.** At the same cash, upside makes people more willing to invest, while lowering how much they save.
- **The wealth gap builds over time.** Higher consumption leaves a smaller financial buffer and less wealth earning investment returns.
- **Cumulative entry remembers the early entrants.** Fewer people first enter in later years than under baseline, roughly offsetting the extra early entrants.

With five years, the willingness boost ends. The smaller buffer remains. With lasting upside, both forces continue.

◀ Back ▶ Entry at the same cash ▶ Annual entry ▶ All diagnostics

What the model explains and what it misses

- Standard saving and costly participation explain why earnings composition can matter.
- Measured cell dynamics strengthen the composition association, mainly through realized resources.
- The coefficient remains much smaller than in the data. Cell differences are useful inputs, not a necessary mechanism.
- Nonholders accumulate too much wealth; former holders re-enter too slowly.

Illustration: more upside can encourage entry now but reduce the saving that finances entry later.

▶ Law and policy accounting ▶ Financial discrepancies ▶ Completed alternatives

Upside earnings opportunities and stock-market entry

- **Finding:** upside earnings opportunities predict stock-market entry at similar total variance and measured resources.
- **Magnitude:** 1.60 percentage points per source SD of composition, about 16 more entrants per 1,000 initial nonholders.
- **Contribution:** separating upside from downside reveals economically important entry differences beyond total variance.
- **Model:** standard saving and participation choices connect earnings composition to entry. Measured earnings dynamics add a positive association, with a large magnitude gap.

▶ Closest empirical papers ▶ Relation to skewness ▶ Discussion appendix

Discussion questions

Empirical interpretation

- ▶ Which literature uses variance?
- ▶ Is this different from skewness?
- ▶ Why education and sex controls?
- ▶ Significant terms, small local variance slope?
- ▶ What is one standard deviation?
- ▶ Which ownership sequences enter?
- ▶ What happens to exits?
- ▶ Why do the samples change?
- ▶ Inference and measurement
- ▶ PSID and period comparisons

Model interpretation

- ▶ How do 58 cells enter the model?
- ▶ Realizations or expectations?
- ▶ What changes under pooling?
- ▶ Why regress on probabilities?
- ▶ Financial fit
- ▶ Why not 580 earnings cells?
- ▶ Entry opportunities and costs
- ▶ Earnings-law and policy accounting
- ▶ Completed magnitude experiments
- ▶ Windfall comparison

◀ Back

Earlier evidence on income risk and portfolios

Study	Empirical risk measure
Guiso, Jappelli & Terlizzese (1996, AER)	Subjective income uncertainty
Angerer & Lam (2009, JF)	Permanent and transitory income variability
Fagereng, Guiso & Pistaferri (2018, ReStud)	Firm-transmitted uninsurable wage risk

This paper studies the balance of above- and below-mean earnings variation and subsequent entry.

Different income-risk measures and portfolio outcomes imply different empirical comparisons.

◀ Back ▶ Theory and measurement

Earnings growth

1. Annual nominal wage growth

$$g_{i,t+1} = \frac{2(y_{i,t+1} - y_{it})}{y_{i,t+1} + y_{it}}, \qquad y_{it} > 0, \quad y_{i,t+1} \geq 0.$$

y_{it}: annual wages of CPS worker i in earnings year t. Zero destination wages: $g_{i,t+1} = -2$.

2. Remove the industry–education age profile

$$\epsilon_{i,t+1} = g_{i,t+1} - \left(\widehat{\theta}_{0G} + \widehat{\theta}_{1G} z_{it} + \widehat{\theta}_{2G} z_{it}^2 + \widehat{\theta}_{3G} z_{it}^3\right).$$

CPS-weighted cubic within industry–education group G.
$z_{it} = (a_{it} - 45)/20$; a_{it} is age at the first interview.

3. Center within the earnings cell

$$\mu_c = \mathbb{E}_c[\epsilon], \qquad \text{within-cell deviation} = \epsilon_{i,t+1} - \mu_c.$$

c: industry–education–age-band cell.

Retirement contributions

Initial non-IRA stock ownership	Composition (pp)	SE
Nonholders	0.22	0.33
Holders	6.36***	2.19
Holder minus nonholder	6.14**	2.33

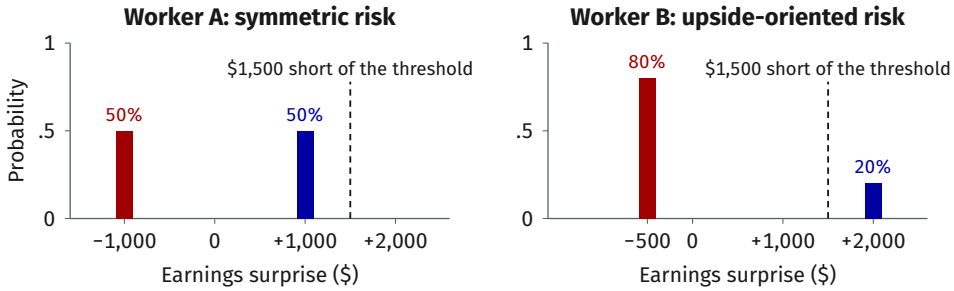
Outcome: any private IRA or annuity contribution

Household expenditure

Asinh dollar scale	Furnishings/ equipment $1,000	Selected nondurables/ services $10,000	Food $10,000
Wage composition, holding variance fixed			
Initial nonholders	0.007	0.001	0.002
	(0.013)	(0.008)	(0.005)
Initial holders	0.114***	0.012	0.009
	(0.041)	(0.020)	(0.017)
Holder minus nonholder	0.107**	0.011	0.006
	(0.046)	(0.024)	(0.019)
Baseline-weight-share average	0.027**	0.003	0.004
	(0.012)	(0.006)	(0.004)
Dispersion, holding composition fixed			
Initial holders	0.021***	0.003	0.001
	(0.006)	(0.004)	(0.003)
33,353 observations; 30 industry–education clusters			

Furnishings: stronger association among initial holders

Earnings gains near the threshold



Entry probability: **0%** for A **20%** for B

Earnings risk and stock market participation

Saving, entry opportunities and holding costs

Budget and portfolio constraints

$$c_t + A_t + D_t \kappa_R \xi_t = M_t,$$

$$M_{t+1} = A_t\{R_{f,t+1} + \alpha_t(R^s_{t+1} - R_{f,t+1})\} + Y_{t+1}.$$

- Participation: $D_t \in \{0, 1\}$, with $D_t \leq a_t$.
- $0 \leq \alpha_t \leq D_t$; no borrowing or short positions.
- Funded equity: $\alpha_t A_t \geq \$1 \times D_t$.

Entry opportunities and holding costs

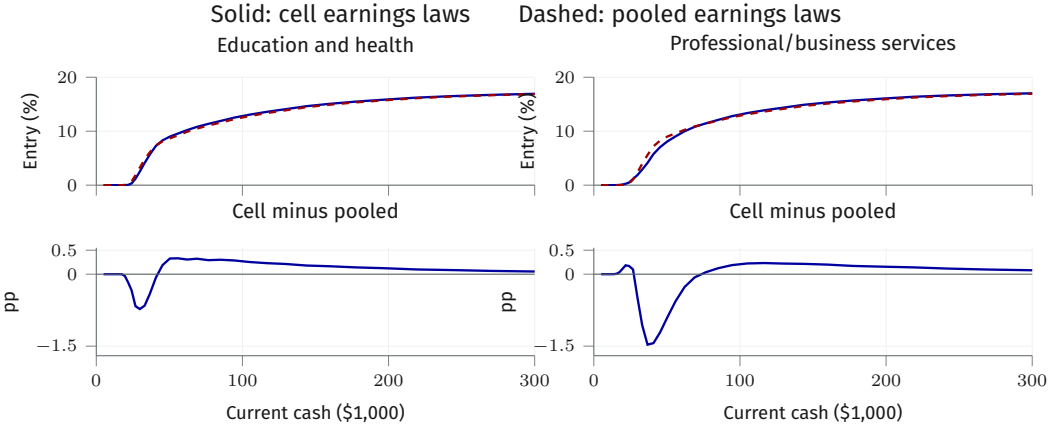
$$\Pr(a_t = 1 \mid D_{t-1}) = D_{t-1} + (1 - D_{t-1})q, \qquad q = 0.176.$$

$$K_t = \kappa_R \xi_t, \qquad \mathbb{E}[\xi_t] = 1, \qquad \kappa_R = \$524.$$

- Nonholders: a new opportunity draw each year.
- Holders: annual choice to continue or exit.
- Access permits entry; payment and investment remain choices.
- After exit: return to the nonholder opportunity process.

▶ Interpretation and selected payments ▶ Friction experiments ▶ Permanent alternative

Entry probabilities by current cash



Targeted financial moments

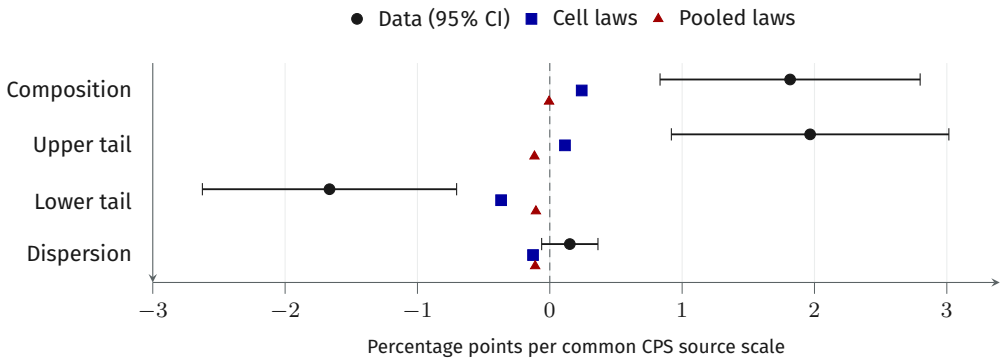
Moment	Unit	Data	SE	Model
Entry: low initial wealth	%	2.93	(0.26)	3.18
Entry: middle initial wealth	%	5.84	(0.36)	5.51
Entry: high initial wealth	%	10.36	(0.48)	10.44
Retention: low initial equity share	%	66.19	(1.43)	67.75
Retention: high initial equity share	%	76.31	(1.23)	74.34
Next equity share: all initial holders	%	33.38	(0.84)	34.40
Wealth change: initial nonholders	$	3,939	(778)	9,941
Wealth change: initial holders	$	1,946	(5,177)	511

Main discrepancy: excess nonholder wealth growth

◀ Back ▶ Additional diagnostics ▶ Alternative fit

Entry estimates from the model

17,606 matched initial nonholders, untargeted coefficients



◀ Back ▶ Permanent alternative

Earnings distributions in portfolio models

Study	Existing mechanism or comparison
Cocco, Gomes & Maenhout (2005)	Labor income, borrowing constraints and downside earnings risk
Catherine (2022)	Earnings downside when stock returns are poor
Shen (2024)	Skewed earnings risk, portfolio choice and participation costs
Gálvez & Paz-Pardo (2026)	Rich earnings dynamics and data/model risk-response regressions

This exercise: assess a specific prospective entry relationship after financial estimation.

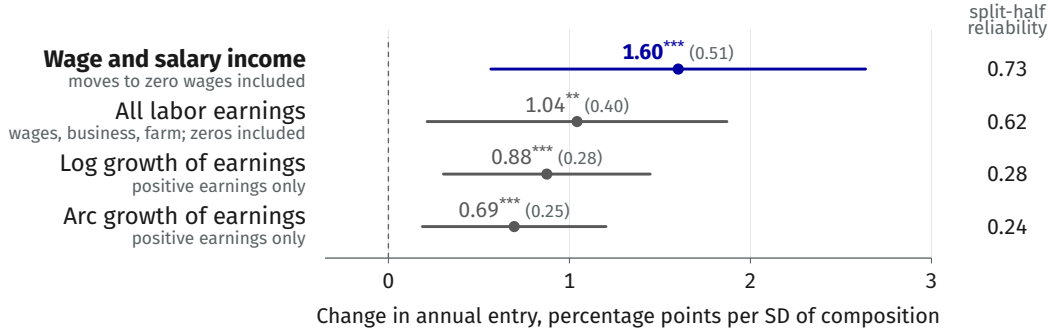
◀ Back ▶ Detailed comparison

Composition at fixed variance

	Symmetric risk	Upside-oriented risk
Lower outcome	$-1{,}000$ (50%)	-500 (80%)
Upper outcome	$+1{,}000$ (50%)	$+2{,}000$ (20%)
Lower partial variance	500,000	200,000
Upper partial variance	500,000	800,000
Total variance	1,000,000	1,000,000
Composition	**0**	**600,000**

Composition: upper minus lower partial variance.

Entry estimates across earnings measures



Positive entry estimates across all four measures. Zero-inclusive wages are the most reliable.

▶ Zeros and growth ▶ Skewness

Observed initial states

21,188 SIPP worker-year pairs, ages 25–63 at origin

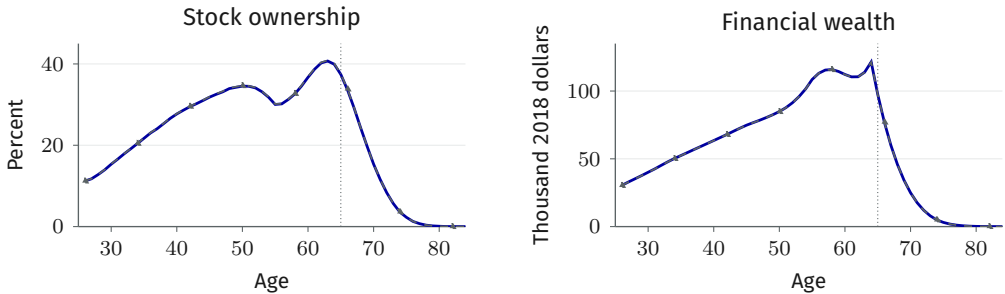
Input	Use in the model
Wages, wealth, ownership and shares	Latent earnings from wages; observed financial positions
Taxes and own pension contributions	Wages net of taxes and contributions
Own retirement income	PSID Social Security and pensions by education and age
Calendar returns	Returns on inherited portfolios in observed transitions

Future return expectations: stationary.

Conditional life-cycle profiles

Conditional projection from 452 observed age-25 origins in 29 groups

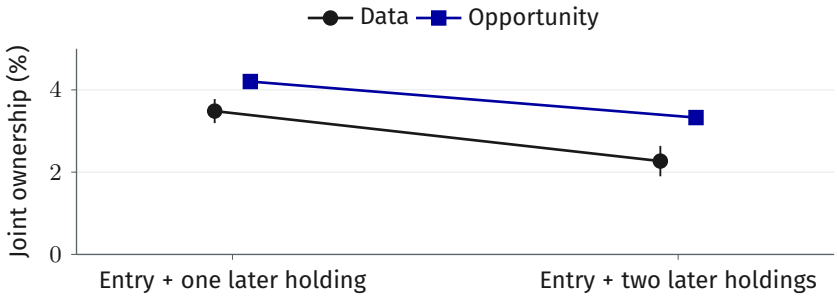
Solid: 141-node grid Dashed triangles: 201-node grid



Numerical accuracy: almost identical paths on finer cash and return grids.

Entry and subsequent ownership

Untargeted predictions on common data/model followup cohorts



Sustained ownership levels remain overpredicted.

▶ Common-cohort estimates

Samples

Sample	Observations
CPS linked earnings histories	88,862
CPS histories with zero destination wages	5,490
SIPP full entry sample	27,419
SIPP prospective allocation sample	31,569
Model financial-fitting worker-year pairs	21,188
Matched model/data entry regression	17,606

Worker transitions, household assets. Mixed-asset mutual funds included. Retirement-account allocation unobserved.

◀ Back ▶ Appendix index

Arc wage growth

$$g_{i,t+1} = \frac{2(y_{i,t+1} - y_{it})}{y_{i,t+1} + y_{it}}.$$

- Positive origin wages, zero destinations included: $g = -2$
- Weighted cubic age adjustment by industry–education group
- Residuals centered within industry–education–age cells
- Zero wages: nonemployment or a different income source, not necessarily dismissal

◀ Back ▶ Appendix index

Relation to skewness

	Composition C	Conventional skewness
Construction	Signed squared deviations	Cubed deviations, divided by SD^3
Interpretation	Upper minus lower contributions to variance	Standardized third moment
Extreme observations	Squared weighting	Cubed weighting

Both describe asymmetry. They are closely related in these data.

Why this measure? Additive variance decomposition; higher measured reliability (0.73 vs. 0.39).

These results do not show that composition predicts entry better than skewness.

◀ Back ▶ Saved skewness comparisons

Composition and skewness: the same controls

Zero-inclusive wage measure	Composition	Skewness
Each measure separately	1.601^{***}	0.912^{**}
	(0.506)	(0.414)
Both measures together	2.585	-0.726
	(1.670)	(1.195)
Split-half reliability	0.73	0.39

Same 27,419 observations; all headline controls, including V and $\sqrt{V}$.

Partial correlation given $V, \sqrt{V}$: 0.91. Neither joint coefficient is precisely separated.

Each measure uses its own source SD. CR1/t_{29} standard errors.

◀ Back ▶ Auxiliary total-earnings comparison

Alternative earnings measures

Construction	Composition (pp)	SE
Wages, zeros included	1.601***	0.506
Total labor earnings, zeros included	1.042**	0.405
Real log growth, positive support	0.875***	0.279
Nominal arc growth, positive support	0.695***	0.248

Controls: $V, \sqrt{V}, \mu, \mu^2$, household characteristics
Industry, education, age-band and year fixed effects

◀ Back ▶ Appendix index

Statistical inference

- **Coefficient tests:** 30 industry–education clusters, CR1/t_{29}
- **Wild-cluster bootstrap:** $p = 0.039$
- **Source uncertainty:** re-estimated CPS moments in separate checks
- Restricted samples: actual cluster count. Unadjusted coefficient stars.

Significant variance terms can give a small local slope

Holding composition and controls fixed, the raw-unit variance terms are

$$f(V) = b_V V + b_S \sqrt{V}, \qquad b_V < 0, \quad b_S > 0.$$

$$f'(V) = b_V + \frac{b_S}{2\sqrt{V}}.$$

A concave relationship: the fitted entry association first rises with variance, flattens, then falls. The two coefficients describe one curve.

Evaluation volatility	Lower quartile	Mean	Upper quartile
Local response (pp)	$+0.438^{***}$	$+0.104$	-0.212
	(0.135)	(0.116)	(0.136)

Full regression including C. Saved local responses per common source SD of C; full-covariance SEs. Raw coefficients are standardized coefficients divided by their respective source SDs. A small slope at the mean does not imply insignificant variance terms.

◀ Back ▶ Appendix index

Risk preferences and sorting

- Risk preferences affect participation thresholds and earnings responses
- Composition slopes may differ across preference groups
- Reported risk appetite: noisy sorting diagnostic

Sorting remains a possible source of the association.

Entry and subsequent ownership

Composition estimate (pp)	One followup	Two followups
Any entry	2.09***	3.31**
Entry and later holdings	0.98*	1.89**
Entry with an interruption	1.11**	1.42
Initial nonholders	9,236	3,254

Joint outcomes partition entry within each cohort.
Imprecise estimates of retention among entrants. Followup selection and weaker three-date reporting screen.

◀ Back ▶ Appendix index

Period differences and age groups

Young-worker SIPP entry gap (pp)	Estimate
Original period gap	2.885
After symmetric overlap weighting	2.623
Change in gap	-0.262
SE of change	0.294
p-value of change	0.381

PSID: all ages, 1999–2017 baselines: -0.18 (SE 0.39).
Ages 45–65, 2009–2017 baselines: 0.44 (SE 1.30). All-age evidence is inconclusive.

Exit among initial holders

A separate risk set: people who already own stocks.

Next-year exit among initial holders	Composition estimate
Exit coefficient	$+3.72$ pp
Clustered standard error	1.71 pp
Wild-cluster bootstrap p-value	0.12

The estimate points toward more exit, but inference is sensitive.

More marginal investors entering and later exiting is one possible explanation. This regression does not test that channel.

▶ Holder outcomes ▶ Selection and interpretation

Initial-holder outcomes

Initial-holder outcome	Composition (pp)	SE
Next-period ownership / retention	-3.72^{**}	1.71
Next risky share, controlling initial share	-0.51	0.94

- Retention wild-cluster bootstrap $p = 0.12$.
- All initial holders, including subsequent exits
- Initial-holder retention differs from survival among new entrants

◀ Back ▶ Appendix index

Participation and consumption

Auxiliary two-date log-utility example with endogenous saving

$$c^*(m, d) = \frac{m - \kappa d}{1 + \beta}.$$

$$\Delta \mathbb{E}[c] = -\frac{\kappa}{1 + \beta} \Delta \Pr(d = 1) \qquad \text{at fixed mean resources.}$$

- Small consumption response possible, exact invariance unnecessary
- IRA/annuity outcome: contribution participation only
- Furnishings: positive primary asinh estimate among holders. Imprecise dollar and original-construction estimates.

The participation threshold follows from optimal saving

Log utility today and tomorrow. Let $b > 0$ be the optimized expected-log-return gain from stock access, independent of earnings.

$$V_d(m) = \max_{0<c<m-\kappa d} \{\log c + \beta[\log(m - \kappa d - c) + bd]\}, \quad d \in \{0, 1\}.$$

$$c_d(m) = \frac{m - \kappa d}{1 + \beta}, \qquad V_1 - V_0 = (1 + \beta) \log\left(1 - \frac{\kappa}{m}\right) + \beta b.$$

$$d^*(m) = 1 \iff m \geq m^* = \frac{\kappa}{1 - \exp[-\beta b/(1 + \beta)]}.$$

The fee absorbs a smaller share of resources when cash is higher.

With no terminal earnings, future resources come from saving. The life-cycle model adds future earnings uncertainty and repeated choices.

◀ Back ▶ Stationary selection formulas

Stationary selection in the fixed-wealth illustration

$$w^* = \frac{\kappa}{1 - e^{-b}} - h, \qquad \pi(w, F) = \Pr(Y \geq w^* - w).$$

$$\text{Entry} = \frac{\mathbb{E}[\pi(1-\pi)]}{\mathbb{E}[1-\pi]}, \qquad \text{Retention} = \frac{\mathbb{E}[\pi^2]}{\mathbb{E}[\pi]},$$

$$\text{Entry + } k \text{ later holdings} = \frac{\mathbb{E}[(1-\pi)\pi^{k+1}]}{\mathbb{E}[1-\pi]}.$$

$b > 0$: optimized log-return gain. h: safe nontradable terminal income.
Fixed wealth, independent draws across dates. Signs depend on resources and earnings laws.

◀ Back ▶ Appendix index

Why 58 cells? Why not 580?

Cells describe earnings environments; more cells require more information.

Twelve industries, three education groups and two age bands give 72 candidates. The common-support screen retains 58.

Support-only age-splitting trial	Current 58	Candidate 580
Median observations per cell	1,284	111
Median effective sample size	787	72
Cells passing existing support rule	58	50

Split each retained cell into ten narrower age groups. About 91% fail the existing support rule.

Finer cells need more data or estimates that share information across groups.

This checks data support only. No new earnings laws or portfolio estimates were fitted.

◀ Back ▶ Model mapping

Full entry projections: composition and local tail contrasts

Upper/lower rows are local algebraic contrasts, not separate regressions.

Entry estimate (pp)	Data	SE	Cell	Pooled	Net
Composition	1.816^{***}	(0.480)	0.241	-0.005	0.247
Upper tail	1.967^{***}	(0.513)	0.115	-0.116	0.230
Lower tail	-1.665^{***}	(0.470)	-0.368	-0.105	-0.263
Dispersion	0.151	(0.104)	-0.127	-0.110	-0.016

- Pooled: both tail estimates negative. Cell: upper positive, lower negative.
- Neither relationship is targeted. Both cell estimates remain much smaller than in the data.

▶ Law and policy contributions ▶ Coefficient figure ▶ Why the test is informative

Selected references

- Catherine (2022). Countercyclical Labor Income Risk and Portfolio Choices over the Life Cycle. *Review of Financial Studies*.
- Catherine, Sodini & Zhang (2024). Countercyclical Income Risk and Portfolio Choices: Evidence from Sweden. *Journal of Finance*.
- Shen (2024). Countercyclical Risks, Consumption, and Portfolio Choice: Theory and Evidence. *Management Science*.
- Bonaparte, Korniotis, Kumar, Michaelides & Zhang (2025). Stock Market Ownership Transitions. *Management Science*.
- Gálvez & Paz-Pardo (2026). Richer earnings dynamics, consumption and portfolio choice over the life cycle. *Journal of Financial Economics*.
- Briggs, Cesarini, Lindqvist & Östling (2021). Windfall gains and stock market participation. *Journal of Financial Economics*.

Closest studies and this paper's contribution

Study	Empirical object
Catherine, Sodini & Zhang (2024)	Current risky shares including zeros, with participation decomposition. Unconditional and cyclical earnings skewness.
Shen (2024)	Changes in risky shares in the stockholder sample, related jointly to skewness and variance.
Bonaparte et al. (2025)	Actual ownership transitions related to income and wealth changes.
Gálvez (2023, working paper)	Participation responses following persistent earnings shocks.

Here: earlier CPS earnings composition and later SIPP entry among initial nonholders, at similar dispersion and resources.

Entry can include former owners returning to the market.

◀ Back ▶ Risk measures ▶ Index

Permanent participation eligibility

$$T_i \in \{0, 1\}, \qquad T_i \text{ remains fixed throughout the path.}$$

- Current holders and reported former holders: $T_i = 1$.
- Among other initial nonholders, $\Pr(T_i = 1) = 0.119$.
- Eligible: annual choice to pay and hold equity. Ineligible: safe asset only.
- Five parameters fitted separately to the same eight financial targets.

Permanent exclusion applies to a fixed share of initially unknown nonholders.

Financial fit under both access models

Moment	Data	Opportunity	Permanent
Entry: low initial wealth	2.93	3.18	3.21
Entry: middle initial wealth	5.84	5.51	5.49
Entry: high initial wealth	10.36	10.44	10.39
Retention: low initial equity share	66.19	67.75	68.01
Retention: high initial equity share	76.31	74.34	73.83
Next equity share: all initial holders	33.38	34.40	33.81
Wealth change: initial nonholders	3,939	9,941	10,005
Wealth change: initial holders	1,946	511	501

Both models: positive wealth growth, excessive nonholder accumulation.

◀ Back ▶ Permanent eligibility ▶ Index

Entry estimates under permanent eligibility

Entry estimate (pp)	Data	Opportunity net	Permanent net
Composition	1.816	0.247	0.171
Upper tail	1.967	0.230	0.155
Lower tail	-1.665	-0.263	-0.188
Dispersion	0.151	-0.016	-0.016

- Composition: pooled 0.211 pp, cell 0.382 pp. A positive association can arise with pooled laws.
- Cells strengthen the association here too. Magnitudes remain well below the data.

Participation dynamics under both access models

Untargeted outcome	Data	Opportunity	Permanent
Re-entry: reported former holders (%)	27.16	9.83	57.68
Entry + one later holding (%)	3.48	4.20	3.82
Entry + two later holdings (%)	2.27	3.33	2.24
Projected age-45 ownership (%)	—	31.83	16.31
Windfall response after two years (pp)	—	15.25	3.44

Permanent eligibility: closer fit to longer sustained ownership; excessive re-entry among reported former holders.

Parameter estimates by access model

	Opportunity	Permanent
Risk aversion γ	8.801	8.917
Discount factor β	0.811	0.810
Recurring mean κ_R	\$524	\$413
Log-cost dispersion σ_κ	0.944	1.013
Opportunity q / active share p	0.176	0.119
EIS ψ, fixed	0.5	0.5
Additional entry fee	0	0

Parameters fixed before earnings-risk and dynamic comparisons. No binding bounds.

Estimation criterion and grid refinement

$$Q = Q_E + 0.1Q_H + Q_W = 9.77.$$

- Marginal covariance blocks: entry, holder outcomes and wealth changes.
- Wealth covariance: $2,000 regularization. Primary-grid growth buffer: $500 for both groups.
- Native convergence. At 201/7: wealth growth of $9,937 for nonholders and $498 for holders.
- Excluded targets: earnings risk, duration, windfalls and life-cycle profiles.

◀ Back ▶ Other numerical diagnostics ▶ Index

Opportunity friction and recurring payments

- Opportunity: a nonholder can reconsider participation.
- Advice, attention and search are possible interpretations, not separately identified.
- Recurring cost paid only when holding equity.
- Offered cost: mean $524, median $335. Participants select lower draws.

Selected recurring payment	Real 2018 dollars
Entrants, supported origins	235
Retained holders	359

Wealth growth and saving net of returns

$$\Delta A_t = \text{gains on inherited assets} + Y_t - c_t - D_t \kappa_R \xi_t.$$

Nonholder wage tercile	Data growth ($)	Model growth ($)
Low	2,388	3,588
Middle	236	8,705
High	9,195	17,535

Holders have positive wealth growth but negative aggregate saving net of returns.

Pooled laws and initial-state sorting

- Fixed: origins, group wage levels, earnings posteriors and calendar returns.
- Remove cell adjustments; reoptimize at common financial parameters.
- Initial resources may remain correlated with earnings cells.
- Pooled holder wealth growth: about $483 below zero. Positive growth is imposed in the fitted cell model.

Four-corner accounting by risk measure

$B(\text{law}, \text{rule})$: next-year outcome law crossed with a previously solved rule.

$$\Delta_L = \tfrac{1}{2}\{B(c,c) - B(p,c) + B(c,p) - B(p,p)\},$$

$$\Delta_P = \tfrac{1}{2}\{B(c,c) - B(c,p) + B(p,c) - B(p,p)\}.$$

Entry estimate (pp)	Laws	Decisions	Total
Composition	+0.274	-0.027	+0.247
Upper tail	+0.302	-0.072	+0.230
Lower tail	-0.245	-0.018	-0.263
Dispersion	+0.029	-0.045	-0.016

Decision rules dampen the upper-tail contribution and reinforce the negative lower-tail contribution.

One-year accounting from common observed origins. The crossed rule retains its own later expectations. Outcomes include cash, work, persistent states and sample selection.

Entry and later ownership: estimates

	One later holding	Two later holdings
Origins	5,459	1,801
Data composition (pp)	0.11	0.77
Data SE	(0.66)	(0.77)
Opportunity net, two seeds	0.19–0.20	0.18–0.22
Permanent net, two seeds	0.14–0.19	0.08–0.13

Smaller common cohorts: imprecise empirical composition estimates. Simulation-seed ranges are not confidence intervals.

Model fit and remaining discrepancies

Comparison	Model result	Discrepancy
Entry by wealth	Increasing across wealth terciles	Imperfect financial fit
Entry and earnings risk	Opposing net tail signs	Magnitudes well below data
Participation dynamics	Entry rises after gains and windfalls	Sustained levels too high; re-entry too low
Saving and positions	Positive wealth changes for both groups	Excess nonholder accumulation; too few small accounts

Permanent alternative: excessive re-entry; closer sustained-ownership fit.

Earnings asymmetry and joint downturn risk

Study	Relevant finding or comparison
Catherine (2022)	Earnings skewness varies with market conditions. Removing that cyclicality preserves earnings variance, persistence and ordinary earnings–return covariance.
Gálvez & Paz-Pardo (2026)	Compares an untargeted income-risk/risky-share regression in data and financially estimated models.
This paper	Upside/downside earnings composition predicts subsequent entry. The model's resources mechanism operates with independent future stock returns.

Composition measures upside-minus-downside partial variance, rather than conventional third-moment skewness or dependence on market returns.

Cell/pooled model predictions change the full earnings law, including mean growth. They do not isolate skewness alone.

◀ Back ▶ Comparison scope

Where the earnings law enters the decision rule

$\rho = 1 - 1/\psi$, $\omega = 1 - \gamma$; x: resources after any recurring payment

$$V^L_{jt}(z,x) = \max_{(A,\alpha)\in\mathcal{A}_j(x)} \left[(x-A)^\rho + \beta s_t \left\{\mathbb{E}_L\left[\bar{V}^L_{t+1}(z', j, AR(\alpha)+Y')^\omega \mid z\right]\right\}^{\rho/\omega}\right]^{1/\rho}.$$

$$W^L_t(z,m,\xi) = \max\{V^L_{0t}(z,m), V^L_{1t}(z,m-\kappa_R\xi)\}.$$

$$\bar{V}^L_t(z,d,m)^\omega = (1-q_d)V^L_{0t}(z,m)^\omega + q_d\mathbb{E}_\xi[W^L_t(z,m,\xi)^\omega], \quad q_d = d + (1-d)q.$$

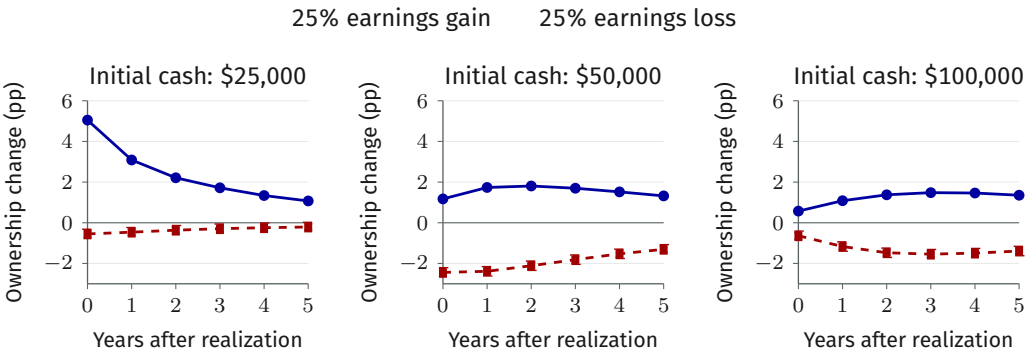
Law L changes future earnings probabilities in $\mathbb{E}_L$ and future values $\bar{V}^L$. The budget and preferences stay the same; the optimal choices can change.

$\mathcal{A}_j$ imposes nonnegative assets and funded equity if $j = 1$. At age 84, $\bar{V}^L_{84}(z,d,m) = m$; no bequest motive.

◀ Back ▶ Timing ▶ Index

Response to realized earnings shocks

Age 45: a one-year shock equal to 25% of a fixed reference gross wage



► Earnings news at fixed current resources ◄ Back

Realized earnings and earnings news

Experiment	What changes	Interpretation
Realized transitory shock	Current cash; fixed future laws	Gains can finance entry
Upper-tail news	Future distribution; fixed current resources	Participation response can be small
Composition news	Future distribution; fixed current resources	Response varies with cash and horizon

Composition-news responses can reverse sign at low cash. More upside need not raise entry.

Numerical precision

- Financial fit, matched entry projections and four-corner accounting: declared finer-grid checks passed.
- Life-cycle grids: 141/5 and 201/7. Some local policies, cash responses and consumption remain grid-sensitive.
- Realized-earnings events: separate 301-node refinement.
- Simulation dispersion and grid differences measure numerical, not sampling, uncertainty.

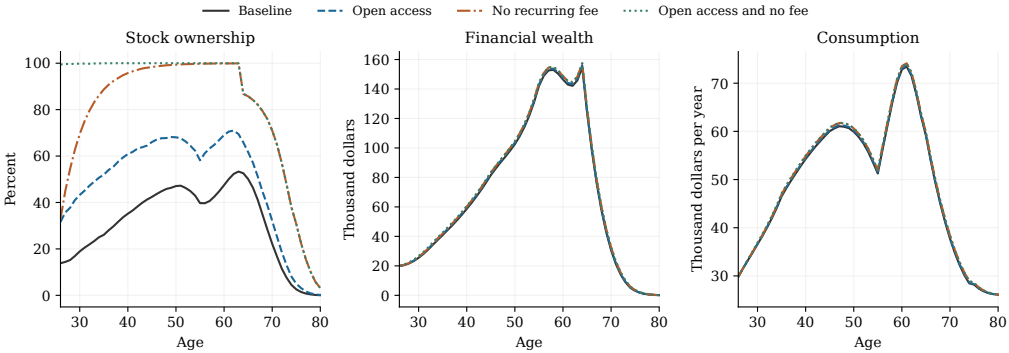
Additional model experiments

Family	Exercises in both specifications
Saving and portfolios	Age profiles, consumption, funded shares, entry and exit
Access frictions	Open access, zero and altered recurring costs
Earnings and retirement	Transitory and persistent risk, income insurance, pensions
Preferences and returns	Risk aversion, patience, EIS, expected returns and volatility
Household responses	Realized earnings, windfalls, fixed-resource news, value of future income

39 scenarios per model, plus conditional policy and valuation diagnostics. Alternative and sensitivity results in the online appendix.

Participation-friction counterfactuals

Remove each friction; reoptimize household choices



SMM targets and weighting

$$\theta = (\gamma, \beta, \kappa_R, \sigma_\kappa, q), \qquad \widehat{\theta} = \arg\min_{\theta} \{Q_E(\theta) + 0.1 Q_H(\theta) + Q_W(\theta)\}.$$

Block	Targets	Economic discipline
Entry (3), Q_E	Three rates by initial wealth	Opportunities and affordability
Holders (3), Q_H	Two retention rates and one next equity share	Continued participation and allocation
Wealth (2), Q_W	Two changes by initial ownership	Saving and resource use

Weight 0.1: entry and wealth growth receive priority over retention and share fit.

◀ Back ▶ Financial fit ▶ All eight targets ▶ Exact criterion

Realized earnings and resources drive the entry pattern

Earnings outcomes: change next year's paycheck and earnings state; apply the same previously solved rule.

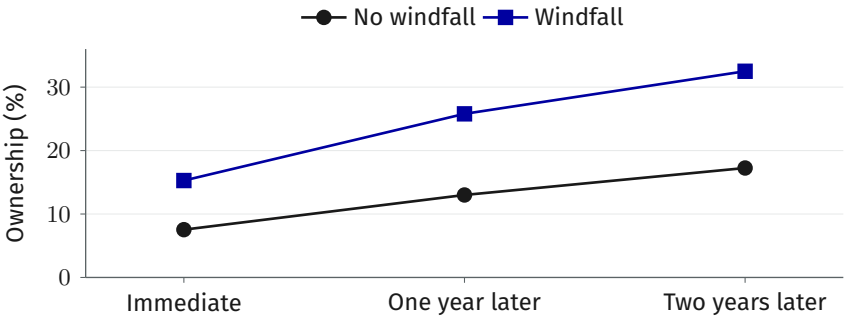
Rule adjustment: keep the next-year earnings law fixed; apply the other plan, solved for different earnings prospects thereafter.

Contribution to the composition estimate	pp per source SD
Earnings outcomes, resources and states	$+0.274$
Reoptimized decision rules	-0.027
Total: cell minus pooled	$+0.247$

The main contribution comes from the outcomes households reach. Rule adjustment provides a modest net offset in this model.

Windfall response

Untargeted experiment: $150,000 to 423 age-44 initial nonholders



Model: $+7.7$ pp immediately; $+15.2$ pp after two years.
Swedish lottery evidence: about $+12$ pp in the winning year (Briggs et al., 2021).

◀ Back ▶ Realized earnings gains ▶ Index

Full entry specification

$$H_{i,t+1} = \alpha + \beta_E \widetilde{C}^{\text{up}}_{c(i,t)} + \gamma_1 \widetilde{V}_{c(i,t)} + \gamma_2 \widetilde{\sqrt{V}}_{c(i,t)} + \gamma_3 \widetilde{\mu}_{c(i,t)} + \gamma_4 \big(\widetilde{\mu}_{c(i,t)}\big)^2$$
$$+ \delta_1 \log\big(\max\{A_{it}, 1\}\big) + \delta_2 \log\big(\max\{\bar{y}^{(1)}_{it} + 1, 1\}\big) + \delta_3 \mathbf{1}\big\{\min_{m \in \mathcal{M}_{it}} y^{(1)}_{itm} \leq 0\big\}$$
$$+ \delta_4 a_{it} + \delta_5 a^2_{it} + \delta_6 \text{Male}_{it} + \eta_{j(i,t)} + \kappa_{e(i,t)} + \lambda_{b(i,t)} + \tau_{t+1} + u_{i,t+1}, \quad H_{it} = 0.$$

- **Outcome:** any-month stock/mutual-fund ownership next year
  Initial nonholders
- **Risk:** composition, dispersion and mean growth
  Fixed CPS source standardization
- **Controls:** assets, earnings, nonpositive earnings and demographics
  Industry, education, age-band and year fixed effects

▶ Inference ▶ Risk preferences

Entry under alternative earnings measures

	(1)	(2)	(3)	(4)
	Wages	Total earnings	Total earnings	Total earnings
	zeros included	zeros included	positive only	positive only
Growth measure	Arc	Arc	Real log	Arc
Upside composition	1.601^{***}	1.042^{**}	0.875^{***}	0.695^{***}
	(0.506)	(0.405)	(0.279)	(0.248)
Composition reliability	0.73	0.62	0.28	0.24
Group-specific cubic age adjustment	Yes	Yes	Yes	Yes
Risk and baseline controls	Yes	Yes	Yes	Yes
Industry, education, age-band, year FE	Yes	Yes	Yes	Yes
Nonholder transitions	27,419	27,419	27,419	27,419
Mean annual entry (%)	8.82	8.82	8.82	8.82
Assignment cells / inference groups	58 / 30	58 / 30	58 / 30	58 / 30

Additional specification results

Specification	Composition (pp)	SE
Prospective-sample baseline	1.65**	0.60
Industry × year; education × year	1.66**	0.62
Industry × education effects	1.86**	0.88
Reported, self-reported, stable households	2.56***	0.56
Controls for realized earnings	1.57**	0.59

▶ Risk preferences and sorting

How the literature uses earnings variance

- Expected-utility models generally depend on the full distribution.
- Empirical studies use variance measures of subjective uncertainty, persistent risk or transitory risk.
- Some calibrated models impose Gaussian earnings innovations; their parameters summarize a restricted distribution.
- Recent portfolio models explicitly allow skewness and nonlinear earnings dynamics.

This paper asks an empirical question: what does the balance of favorable and adverse variation predict about entry, given dispersion?

Which comparisons are performed?

Comparison	Scope
Composition and skewness	Closely related. Saved joint regressions do not establish incremental information beyond skewness.
Cell and pooled earnings	Same model and financial parameters; optimal rules recomputed. This is the main earnings-environment comparison.
Opportunity and permanent access	Separately fitted to the same financial targets; different dynamic misses
Published structural models	Mechanism comparison. No common-data re-estimation of entire published models.

◀ Back ▶ Skewness horse race ▶ Access models

Auxiliary total-earnings horse race

Saved zero-inclusive total-earnings comparison	Result
Composition coefficient (pp)	3.782
Two-survey standard error (pp)	2.261
Composition p-value	0.100
Two-test adjusted q-value	0.200
Joint test: both slopes zero	$p = 0.038$
Joint test: broader dependence sensitivity	$p = 0.145$

Total earnings, including zeros; adds zero-wage probability. Separate two-survey inference.

This is not the headline wage regression. The evidence does not establish an incremental association beyond skewness.

Measurement adjustment and regression controls

Step	Purpose
CPS earnings construction	Remove an industry–education-specific age profile from growth, then calculate within-cell distributional moments.
SIPP entry regression	Condition participation on individual resources, age and sex, plus industry, education, age-band and year fixed effects.

Education appears in both steps because they answer different questions.

Sex is an individual participation control; the headline risk cells are not sex-specific.

Separate industry, education and age-band effects do not absorb every industry–education–age interaction.

◀ Back ▶ Construction ▶ Full regression

The coefficient's unit

$$\widetilde{C}_c = \frac{C_c - \bar{C}_{\mathrm{CPS}}}{s_{C,\mathrm{CPS}}}, \qquad s_{C,\mathrm{CPS}} \simeq 0.034.$$

A one-unit increase in $\widetilde{C}$ means a 0.034 increase in raw composition.

The fitted coefficient is 0.01601 in probability units, or 1.601 percentage points.

It uses the same dispersion and individual controls throughout.

This unit describes differences between earnings environments. It is not a 1% earnings rise or an individual windfall.

◀ Back ▶ Tail units ▶ Illustration

Why different questions use different samples

Question	Observations	Reason
Main SIPP entry	27,419	Initial nonholders with required annual outcomes and controls
Entry plus later holdings	9,236 / 3,254	One / two additional complete followups
Model financial fit	21,188	Matched usable model states, including initial holders
Model entry regression	17,606	Initial nonholders within the model sample
SCE reallocation / share	2,136 / 3,332	Different outcome availability and eligibility

The matched empirical model coefficient is re-evaluated on those same 17,606 origins. Do not compare its model coefficient with the 27,419-origin headline as if the samples were identical.

◀ Back ▶ Additional samples

Which ownership sequences enter the sample?

Initial nonholders with observed ownership in every required follow-up year.

Four-year sequence	Entry	Entry + two later holdings
$0, 1, 1, 1$	1	1
$0, 1, 0, 1$	1	0
$0, 0, 1, 1$	0	0
$1, 0, 1, 1$	Outside the initial-nonholder sample	

Each year defines a new baseline. A later $0, 1, 1$ window can enter the shorter-horizon cohort.

Missing required follow-up excludes the observation. Ownership means any stockholding during that year.

Retention among entrants is a separate outcome; its estimates are imprecise.

Why probabilities do not mechanically shrink slopes

$$\widehat{\beta}(Y) = (X'WX)^{-1}X'WY.$$

If $p = \mathbb{E}[Y \mid X]$, then

$$\mathbb{E}[\widehat{\beta}(Y) \mid X] = \widehat{\beta}(p).$$

Averaging binary realizations into probabilities removes simulation noise; it does not mechanically reduce the expected regression coefficient.

Current matched mean: model 6.375%, data 6.375%. No exact-zero model probabilities among 17,606 origins.

The shortfall concerns how probabilities vary with earnings composition. Model probabilities need not equal the true conditional expectation.

◀ Back ▶ Coefficient comparison

Completed alternatives do not close the magnitude gap

Specification	Composition (pp)	Nonholder wealth error	Holder wealth error
Current cell model	0.241	Reference	Reference
Cash-dependent opportunities	0.337	+$157	+$11
$50 recurring cost + setup	0.267	+$302	+$10
$100 recurring cost + setup	0.253	+$362	+$11
Matched empirical coefficient	1.816		

Wealth columns: change in each group's absolute wealth-growth error; positive is worse. Same 141-node resolution. Both groups also worsen on the finer grid.

Low-fee versions predict retention above 96%. No tested specification resolves magnitude while preserving the requested wealth fit.

◀ Back ▶ Permanent alternative

The earnings law: states and wage levels

$$F^{L}_{g,t}(\eta_{t+1}, n_{t+1}, W_{t+1} \mid \eta_t, n_t), \qquad L \in \{\text{cell}, \text{pooled}\}.$$
$$W_{gt} = \mathbf{1}\{n_t = 0\} \exp\{f_h(t) + \ell_g + B_g(t) + \eta_t + \varepsilon_t\}.$$

State / component	Meaning
n_t	0 with wage work, 1 without wage work
$h,\ g$	Education type, industry–education group
$f_h(t),\ \ell_g$	Education age profile and group wage level
$B_g(t),\ \eta_t,\ \varepsilon_t$	Cell growth profile, persistent state, transitory shock

Gross wages W_{gt} enter disposable resources Y_t after taxes and employee contributions. Other resources can remain when wages are zero.

◀ Back ▶ Cell coefficients ▶ Career transitions ▶ Cell and pooled

The cell adjustment vector and its joint targets

Coefficient	What changes
a_c^{exit}	Wage-work departure hazard
b_c	Deterministic age-growth profile
s_c	Transitory and reporting-error scale
a_c	Transitory probabilities and persistent jump hazards

$$B_g(t) = \begin{cases} b_{g0}(t-35), & t < 45, \\ 10b_{g0} + b_{g1}(t-45), & t \geq 45. \end{cases}$$

Fit the four coefficients jointly to the zero-wage share, mean arc growth, and upper/lower partial variances.

The persistent versus transitory allocation of asymmetry is weakly identified. Pooling sets all four adjustments to zero and reoptimizes choices.

◀ Back ▶ Persistent transitions ▶ Temporary shocks

Persistent earnings and wage-work transitions

$$\eta_{t+1} = \rho_h \eta_t + \zeta_{t+1} + J_h^+ I_{t+1}^+ - J_{h,b}^- I_{t+1}^- - m_{g,b}, \quad \zeta_{t+1} \sim N(0, s_{0h}^2).$$

ρ_h: persistence. $m_{g,b}$: centering drift. Promotion, loss and no jump are mutually exclusive outcomes.

$$\widetilde{p}_{c,b}^{\pm} = \min\{\widehat{p}_{c,b}^{\pm}, .45\}, \qquad p_{c,b}^{\pm}(\eta) = \min\{\widetilde{p}_{c,b}^{\pm} e^{\pm k_h^{\pm} \eta/\sigma_{\eta,h}}, .45\}.$$

$\widehat{p}^{\pm}$ are the tilted probabilities before caps, defined on the linked normalization slide.

$$\Pr(n_{t+1} = 1 \mid n_t = 0, \eta_t) = \min\{\delta_{h,b} e^{a_c^{\text{exit}}} e^{-\phi_h \eta_t/\bar{\eta}_h}, .6\},$$

$$\Pr(n_{t+1} = 0 \mid n_t = 1) = \min\{r_{h,b}^0 e^{r_h}, .95\}.$$

$n = 0$: wage work. $n = 1$: no wage work. Earnings states evolve in both states. Wage-work return r_h differs from investment opportunity q.

◀ Back ▶ Tilt and normalization ▶ Parameters

What does the persistent-shock normalization hold fixed?

Before caps and state dependence, cell c tilts promotion/loss probabilities:

$$\widehat{p}^{+}_{c,b} = \chi_{c,b} p^{+}_{h,b} e^{a_c}, \qquad \widehat{p}^{-}_{c,b} = \chi_{c,b} p^{-}_{h,b} e^{-a_c},$$

$$\chi_{c,b} = \frac{p^{+}_{h,b}(J^{+}_{h})^2 + p^{-}_{h,b}(J^{-}_{h,b})^2}{p^{+}_{h,b} e^{a_c}(J^{+}_{h})^2 + p^{-}_{h,b} e^{-a_c}(J^{-}_{h,b})^2}.$$

$$\widehat{p}^{+}_{c,b}(J^{+}_{h})^2 + \widehat{p}^{-}_{c,b}(J^{-}_{h,b})^2 = p^{+}_{h,b}(J^{+}_{h})^2 + p^{-}_{h,b}(J^{-}_{h,b})^2.$$

This fixes a raw jump second moment before later adjustments.

Probability caps and state dependence follow. Centering, employment transitions and temporary shocks also matter for total earnings variance and growth.

◀ Back ▶ Complete transition ▶ Temporary component

Transitory earnings and reporting error

Transitory earnings

$$\varepsilon^0 \sim (1-p_2)N(0,s_1^2) + p_2 N(m_2, s_2^2).$$

Ten raw nodes e_j with probabilities p_{0j}. Cell tilt and scale give

$$p_{cj} \propto p_{0j} \exp\{a_c \operatorname{sgn}(x_j) x_j^2\}, \quad x_j = \operatorname{clip}\big((e_j - E_0 e)/sd_0(e), -4, 4\big),$$

$$\varepsilon_{cj} = e^{s_c} e_j - \log \sum_k p_{ck} e^{e^{s_c} e_k} + \log \sum_k p_{0k} e^{e_k}.$$

The location adjustment preserves $E[e^{\varepsilon}]$, the baseline wage multiplier. It does not fix variance.

Reporting error

$$\log W_t^{\text{obs}} = \log W_t + u_t \quad (W_t > 0), \qquad u_t^0 \sim (1-p_u)N(0,t_1^2) + p_u N(0,t_2^2).$$

Cell scale also multiplies reporting-error nodes by e^{s_c}. Reporting error changes measurement, not household resources.

◀ Back ▶ Shock parameters ▶ Cell coefficients

Earnings parameters: career dynamics

Parameter	Noncollege	College
Persistence ρ_h	0.946	0.944
Small innovation SD s_{0h}	0.024	0.037
Promotion size J_h^+	0.328	0.522
Loss size J_h^-, ages 25–54	0.742	0.419
Loss size J_h^-, ages 55–64	1.150	1.346
Promotion state sensitivity k_h^+	0.248	0.140
Loss state sensitivity k_h^-	0.148	0.931
Age-band order: 25–34, 35–44, 45–54, 55–64		
Base promotion probabilities $p_{h,b}^+$	.117, .039, .013, .046	.082, .019, .009, .021
Base loss probabilities $p_{h,b}^-$	.036, .032, .023, .035	.010, .029, .030, .043
$f_h(t) = a_{0h} + a_{1h}x + a_{2h}x^2 + a_{3h}x^3, \quad x = (t-45)/20$		
a_{0h}, a_{1h}	10.502, .202	11.153, .118
a_{2h}, a_{3h}	$-.239, -.004$	$-.361, .087$

Jump sizes in log points; annual probabilities before cell and state adjustments.

◀ Back ▶ Other parameters ▶ Main earnings slide

Earnings parameters: shocks and wage work

Parameter	Noncollege	College
Transitory large-component probability p_2	.044	.031
Transitory small-component SD s_1	.060	.101
Transitory large-component mean m_2	$-.339$	$-.349$
Transitory large-component SD s_2	.627	.620
Reporting large-component probability p_u	.230	.231
Reporting small-component SD t_1	.164	.182
Reporting large-component SD t_2	.913	.819
Departure state sensitivity ϕ_h	1.463	1.435
Wage-work return log shift r_h	1.926	1.996
Age-band order: 25–34, 35–44, 45–54, 55–64		
Base departure probabilities $\delta_{h,b}$	.074, .068, .057, .073	.039, .030, .041, .069
Return inputs $r^0_{h,b}$	.244, .170, .084, .047	.924, .292, .263, .094

Log-point shocks. Return inputs precede the fitted shift and probability cap.

◀ Back ▶ Career parameters ▶ Law definition ▶ Main

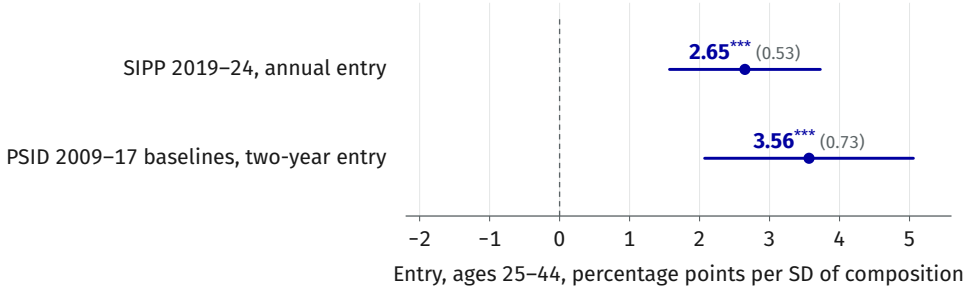
Why measure earnings risk earlier?

- CPS earnings changes are measured before the SIPP investment outcomes.
- A current shock cannot directly enter both the historical risk measure and the current portfolio report.
- Persistent group conditions, worker sorting and correlated shocks can still connect the two periods.

The timing reduces a source of contemporaneous correlation. It does not establish causal assignment of earnings risk.

◀ Back ▶ Group controls

Entry at ages 25–44



▶ Period gap and all-age results

Why earnings targets do not guarantee entry signs

1. Fit wages and wage-work transitions to earnings data.
2. Financial targets discipline saving, participation costs and access.
3. Neither step targets how entry varies with earnings composition.

The model must convert those earnings distributions into the observed entry response through resources and choices.

Matching the earnings moments alone does not guarantee the sign or size of that response.

How large is the entry difference?

Study and outcome	Risk comparison	Difference
This paper: next-year entry among nonholders	1 source SD in composition	$+1.60$ pp
Catherine, Sodini & Zhang (2024): current participation	2 SD in cyclical skewness	-2.3 to -8.5 pp

Here: 1.60 points relative to an 8.82% mean entry rate.

About 16 more entrants per 1,000; 18% of the mean, not 18 percentage points.

Different risk measures, populations and outcomes. The published range is across specifications, not a confidence interval.

Source: Catherine, Sodini & Zhang (2024), Table IV and p. 1774.

Related work in progress

van den Akker, de Bresser, Teppa & Trucchi (2026)

Subjective Expectations and Household Portfolio Choices

- Subjective expectations linked to administrative portfolios.
- Abstract: upward income risk predicts investment; entry patterns are concentrated among younger households.

Here: measured group earnings composition and entry, controlling for total variance and resources.

Related conference work, verified in the IAAE 2026 program. Full specifications and journal publication have not been verified.

Two-period illustration: the optimization problem

Choose consumption c_0, safe saving s, stock investment x and entry $d \in \{0, 1\}$:

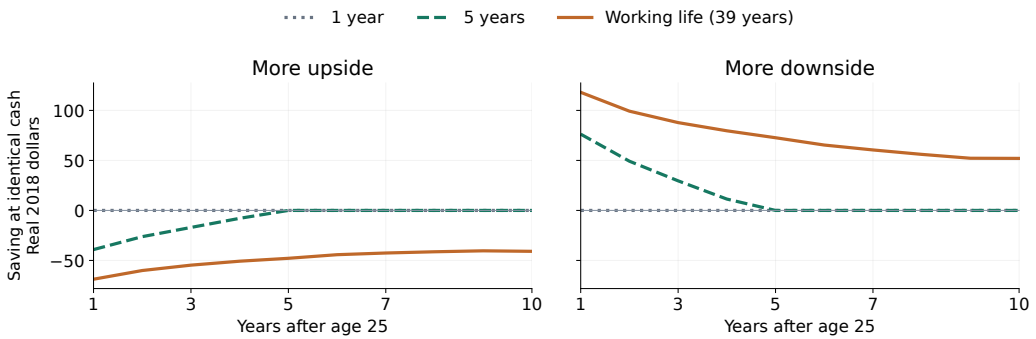
$$\max_{s,x,d} \log c_0 + \mathbb{E} \log c_1, \qquad c_0 = m - s - x - 1.5d, \qquad c_1 = Y_1 + s + Rx.$$

- Same log utility and discount factor $\beta = 1$. No borrowing or shorting: $s, x \geq 0$, with $x = 0$ if $d = 0$ and $c_0 > 0$.
- Safe gross return is 1. Stock gross return R is 0.5 or 1.7, each with probability one half, independent of earnings.
- Y_1 follows U or D. In the realized-resource panel, $m = Y_0$ with zero inherited assets; Y_0 and Y_1 are independent draws from the same law.
- Financial saving is $s + x$: it excludes entry fees and subsequent returns. All amounts are in thousands of dollars.

Optimize with and without entry, then choose the higher value. These are teaching assumptions, not estimated parameters of the life-cycle model.

◀ Back ▶ Expectations ▶ Resources

At the same cash, how does duration affect saving?



Lasting upside lowers saving even in the no-stock alternative.
At the same cash: $69 less in year 1 and $41 less in year 10. The one-year response is zero because future earnings are already baseline at the first new choice.

◀ Back ▶ Entry at the same cash ▶ All diagnostics

Which comparison answers which question?

How do later first-entry rates change? Annual first entry

Do choices change at the same cash? Entry decisions Saving

Earnings paths, decision rules and the household budget

Duration	Entry	Wealth	Budget
One year	Decomposition	Decomposition	Accounting
Five years	Decomposition	Decomposition	Accounting
Working life	Decomposition	Decomposition	Accounting

All comparisons use the same initial nonholders and fixed parameters. The shocks illustrate mechanisms. Earnings-risk coefficients are untargeted.

◀ Back ▶ Experiment design

Lasting upside: fewer first-time entrants in later years



Fewer people first enter from year 3 onward under lasting upside.
The early entry lead shrinks. Cumulative first entry still counts those who entered earlier.

◀ Back ▶ Cumulative entry ▶ Same-cash decisions

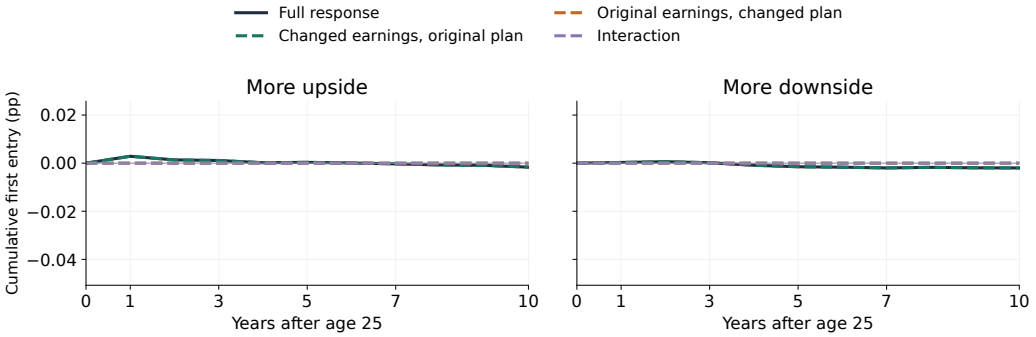
Lasting upside keeps entry more attractive at the same cash



Lasting upside: the entry-decision boost remains in year 10.
Current nonholders on baseline paths, at identical resources and states. This includes re-entry. Probe choices do not alter subsequent states.

◀ Back ▶ Same-cash saving ▶ All diagnostics

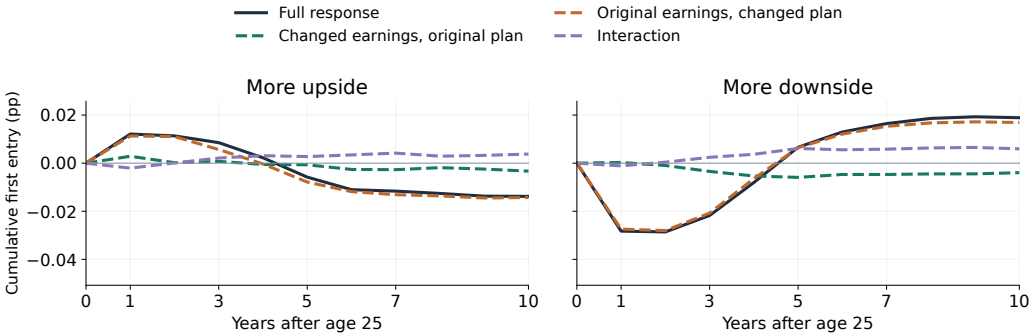
One year: earnings paths and entry



Only realized resources change at the first new decision.
The one-year change has ended for all future earnings. Rule-only and interaction responses are exactly zero. Tiny entry signs are grid-sensitive.

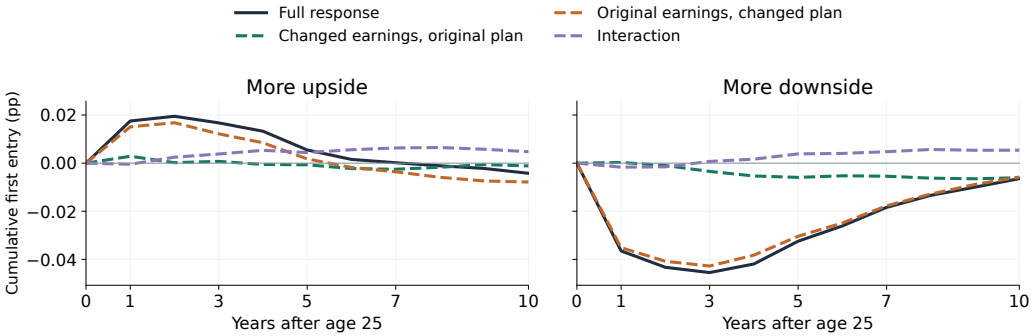
◀ Back ▶ Entry ▶ Wealth ▶ Budget

Five years: earnings paths and entry



The changed plan drives most of the later cumulative-entry reversal.
The crossed rule comparison includes saving and the resources it creates over time. It is not an equal-cash entry probe.

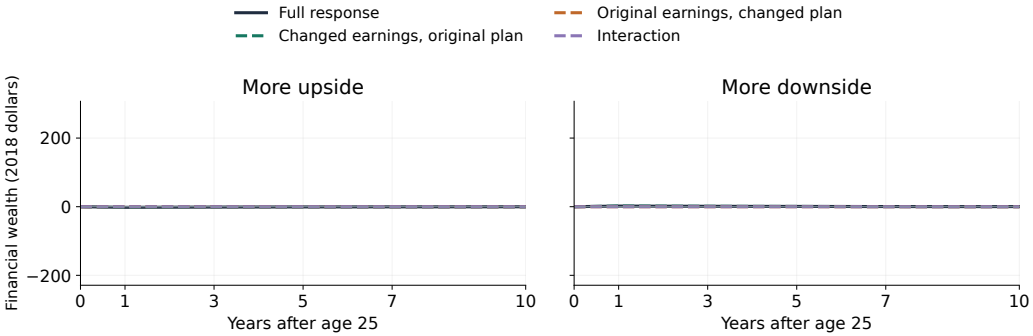
Working life: earnings paths and entry



The full cumulative-entry response ends close to zero.
Changed earnings and changed plans interact. The changed plan includes saving, investment and the later resources they produce.

◀ Back ▶ Entry ▶ Wealth ▶ Budget

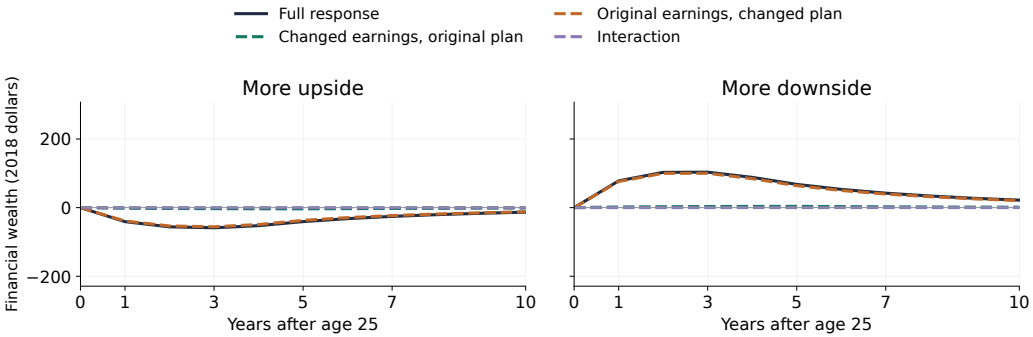
One year: earnings paths and wealth



The small wealth effect comes through realized resources.
Future earnings prospects are baseline at the first new choice. Households use the original rule at their changed cash.

◀ Back ▶ Entry ▶ Wealth ▶ Budget

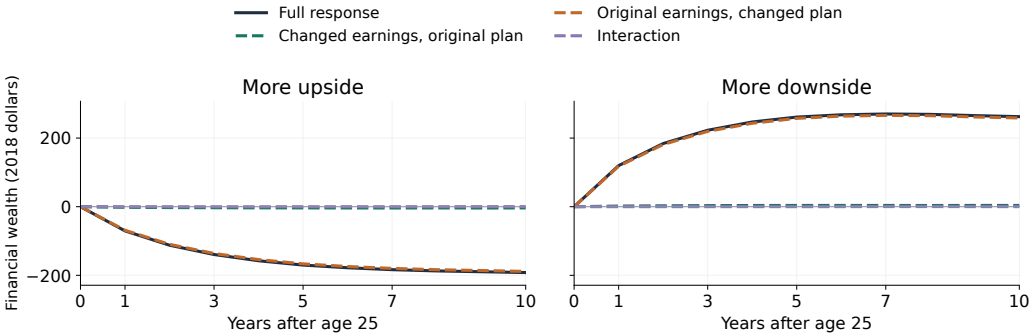
Five years: earnings paths and wealth



Earlier choices leave a wealth gap after the regime ends.
Rules coincide at the same state from year 5. The financial resources reached along each path can still differ.

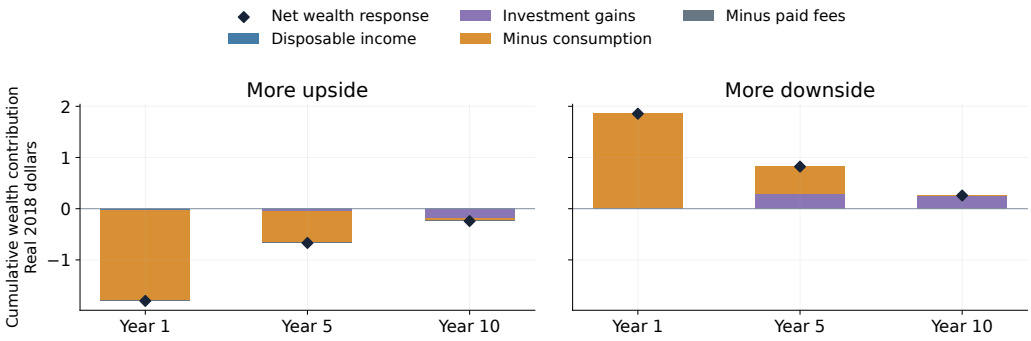
◀ Back ▶ Entry ▶ Wealth ▶ Budget

Working life: earnings paths and wealth



The changed plan accounts for most of the persistent wealth gap.
At year 10 under upside, the plan-only contrast is about -$189. Changed earnings with the original plan contribute about -$4.

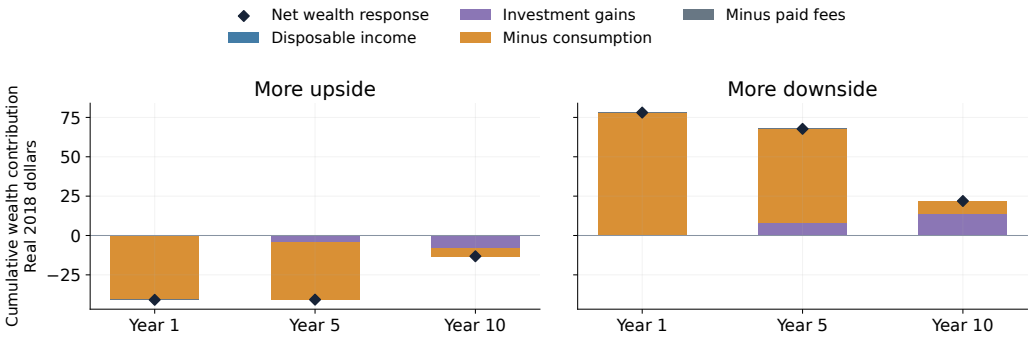
One year: where does the wealth difference come from?



With one year, the budget response is small.
After current earnings arrive, future prospects are already baseline. The current income draw changes resources, consumption and saving.

◀ Back ▶ Wealth decomposition ▶ Saving at the same cash

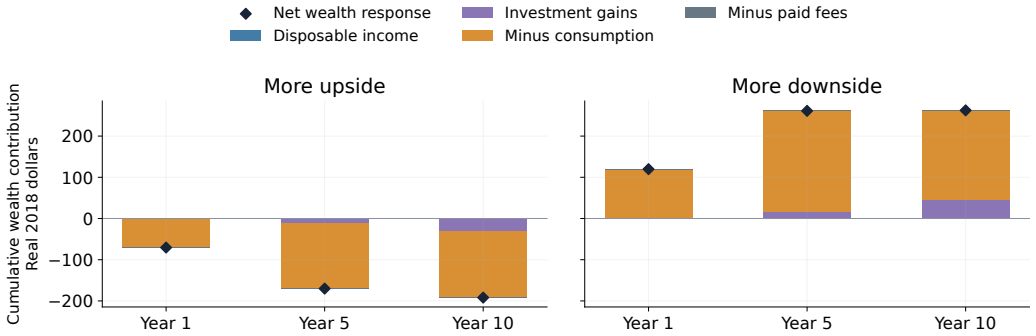
Five years: where does the wealth difference come from?



Consumption explains most of the initial wealth response.
As the temporary change ends, spending responses diminish. Earlier saving and investment choices still affect wealth.

◀ Back ▶ Wealth decomposition ▶ Saving at the same cash

Working life: where does the wealth difference come from?



Higher consumption accounts for most of the upside wealth decline.
At year 10: about -$163 from consumption, -$28 from investment gains and -$0.18 from fees.

◄ Back ► Wealth decomposition ► Saving at the same cash